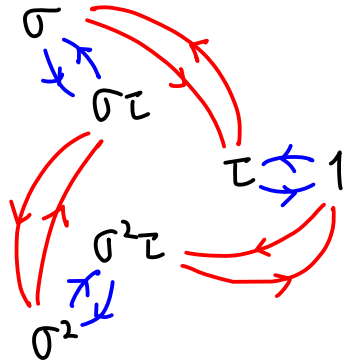
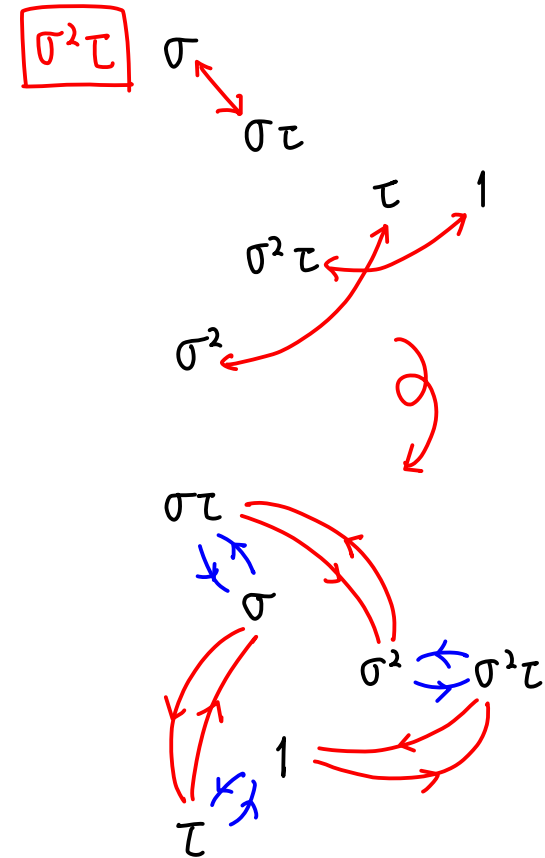
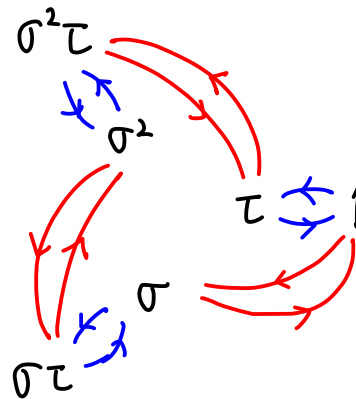
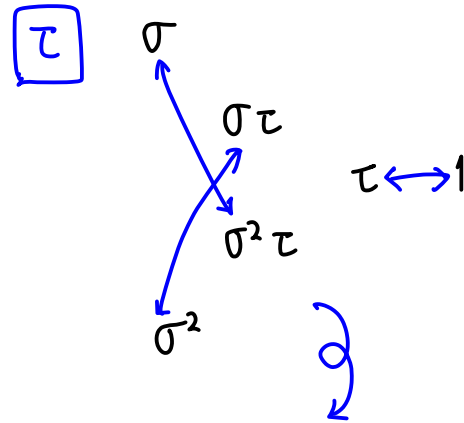


S_3 における $\tau = (1, 2)$ と $\tau\sigma = \sigma^2\tau = (2, 3)$ の

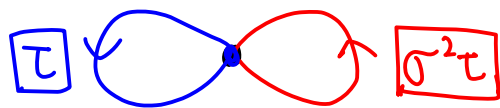
右作用に関する Cayley グラフ



Cayley グラフへの左作用

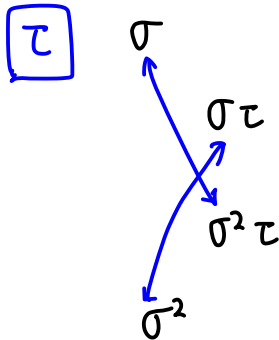
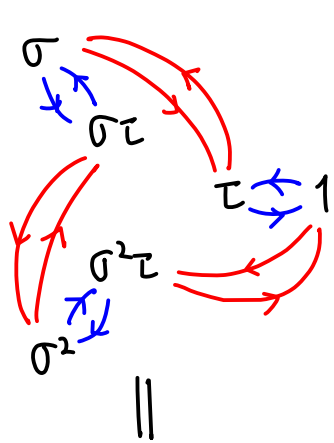


上の Cayley グラフは 8 の字の S_3 -Galois 被覆になる.

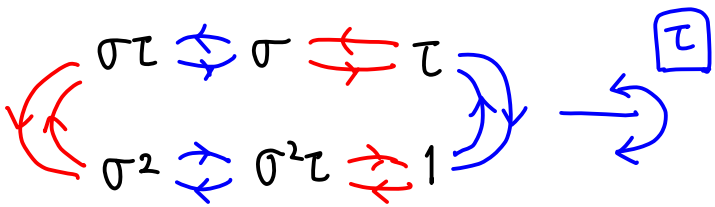
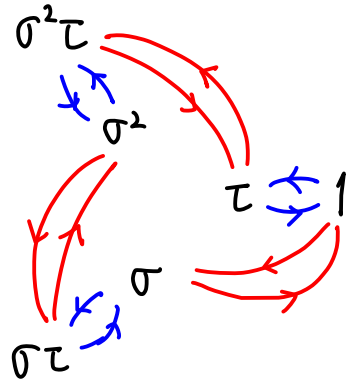


S_3 における $\tau = (1, 2)$ と $\tau\sigma = \sigma^2\tau = (2, 3)$ の

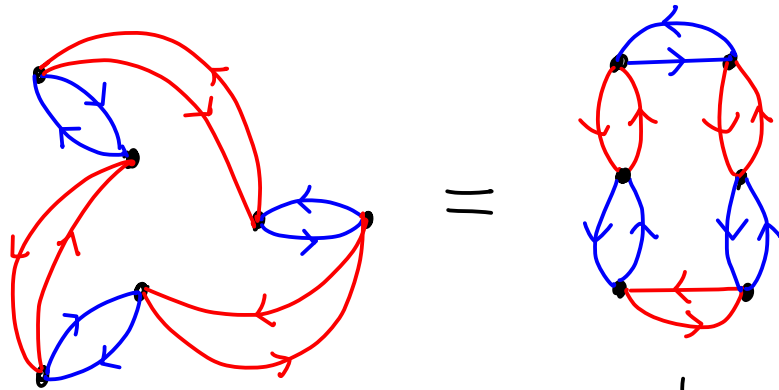
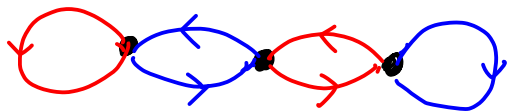
右作用に関する Cayley グラフ も Cayley グラフへの $\langle \tau \rangle$ 左作用で割る.



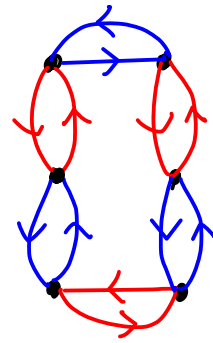
$1 \leftrightarrow \tau$



$\langle \tau \rangle$ で割る



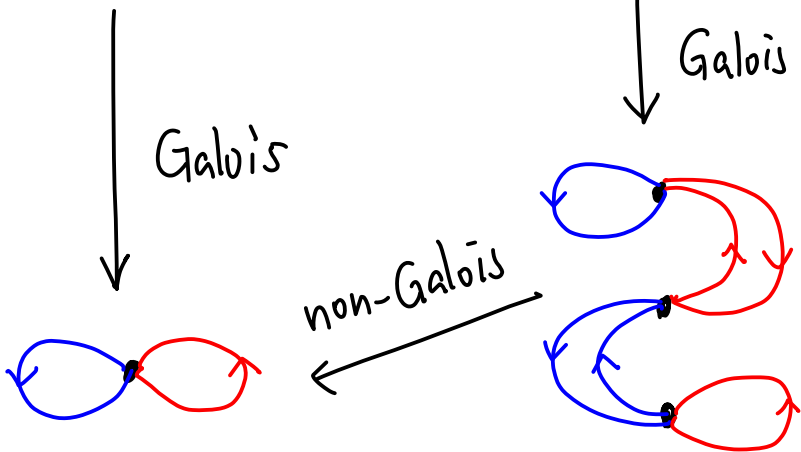
$=$



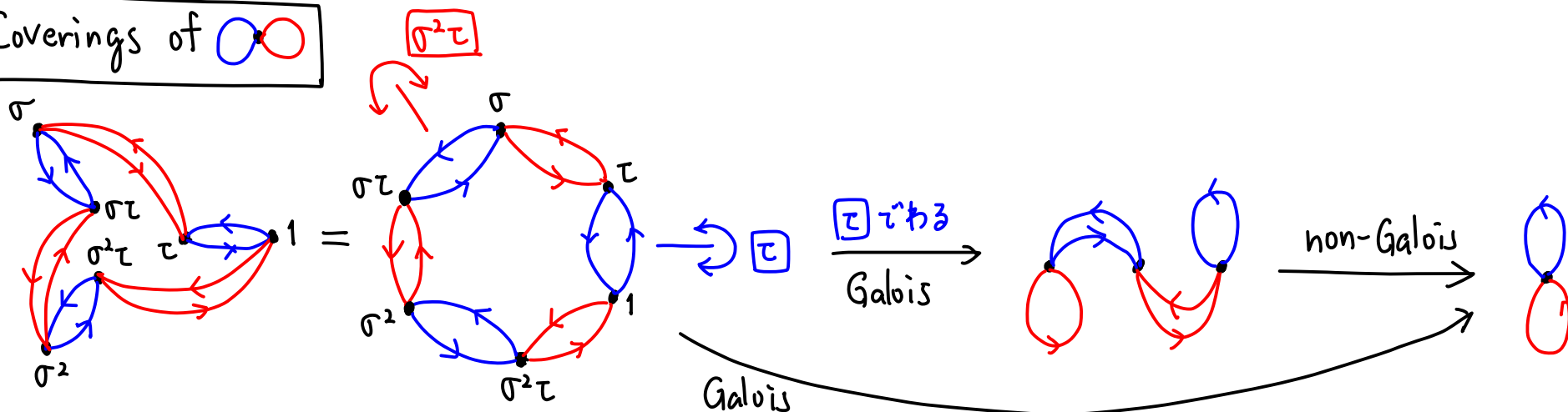
Galois

Galois

non-Galois



Coverings of



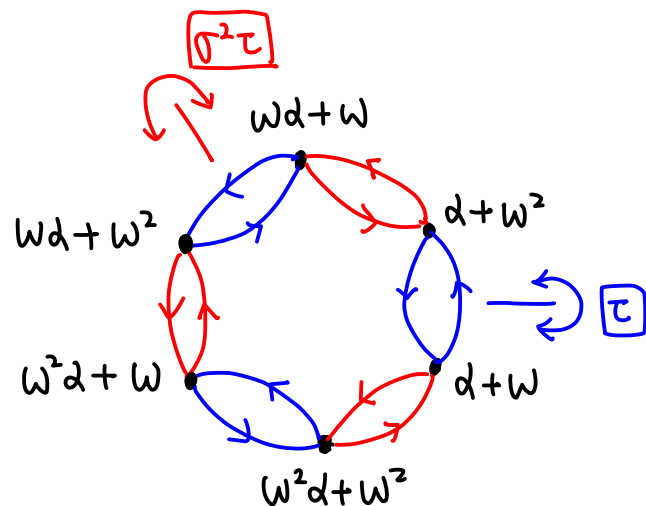
Extensions of $\mathbb{Q}$

$$\alpha = \sqrt[3]{2}, \quad \omega = \frac{-1 + \sqrt{-3}}{2} \text{ とおく.}$$

$$\mathbb{Q}(\alpha + \omega) = \mathbb{Q}(\alpha, \omega) \xleftarrow{\text{Galois}} \mathbb{Q}(\alpha) \xleftarrow{\text{non-Galois}} \mathbb{Q}$$

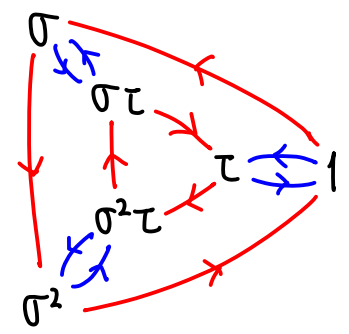
$\xleftarrow{\text{Galois}}$

$$\sigma(\alpha) = \omega\alpha, \quad \sigma(\omega) = \omega, \quad \tau(\alpha) = \alpha, \quad \tau(\omega) = \omega^2 \text{ とする.} \quad \sigma^2\tau(\alpha) = \omega^2\alpha, \quad \sigma^2\tau(\omega) = \omega^2, \quad \sigma^2\tau(\omega\alpha) = \omega\alpha.$$

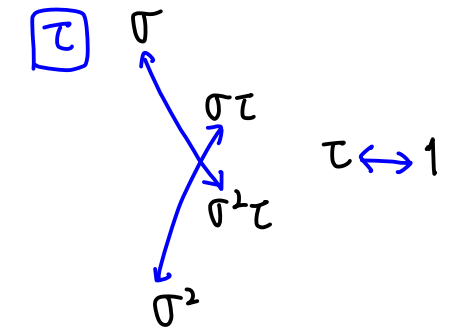
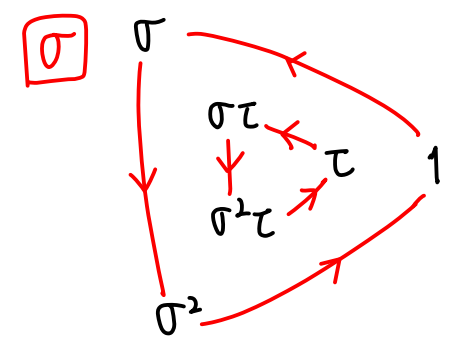


S_3 における $\tau = (1, 2)$ と $\sigma = (1, 2, 3)$ の

右作用に関する Cayley グラフ

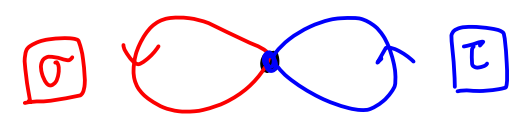


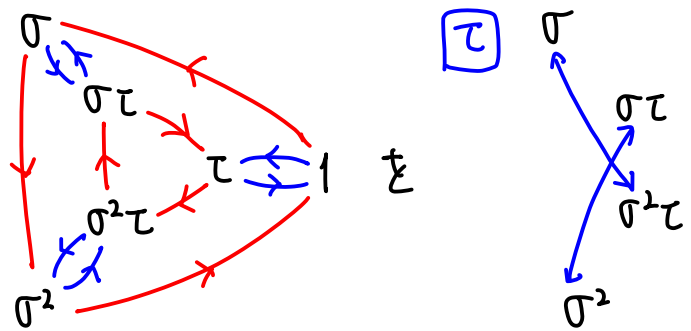
Cayley グラフへの左作用



これらの作用は Cayley グラフの色と向きを保つ。

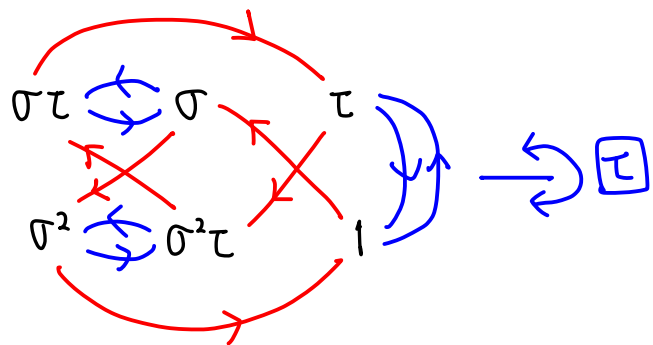
上の Cayley グラフは 8 の字の S_3 -Galois 被覆になる。



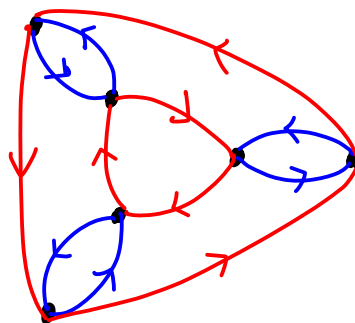
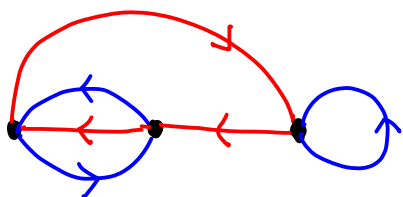


$\tau \leftrightarrow 1$ $\tau^2 \leftrightarrow 2$

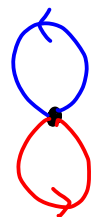
||



2で割る

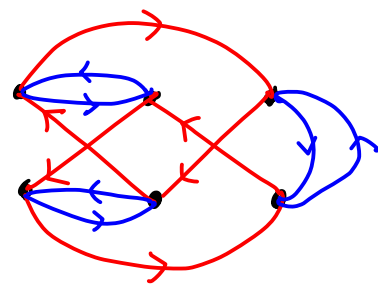


Galois

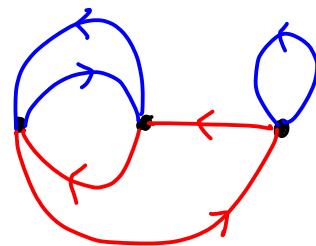


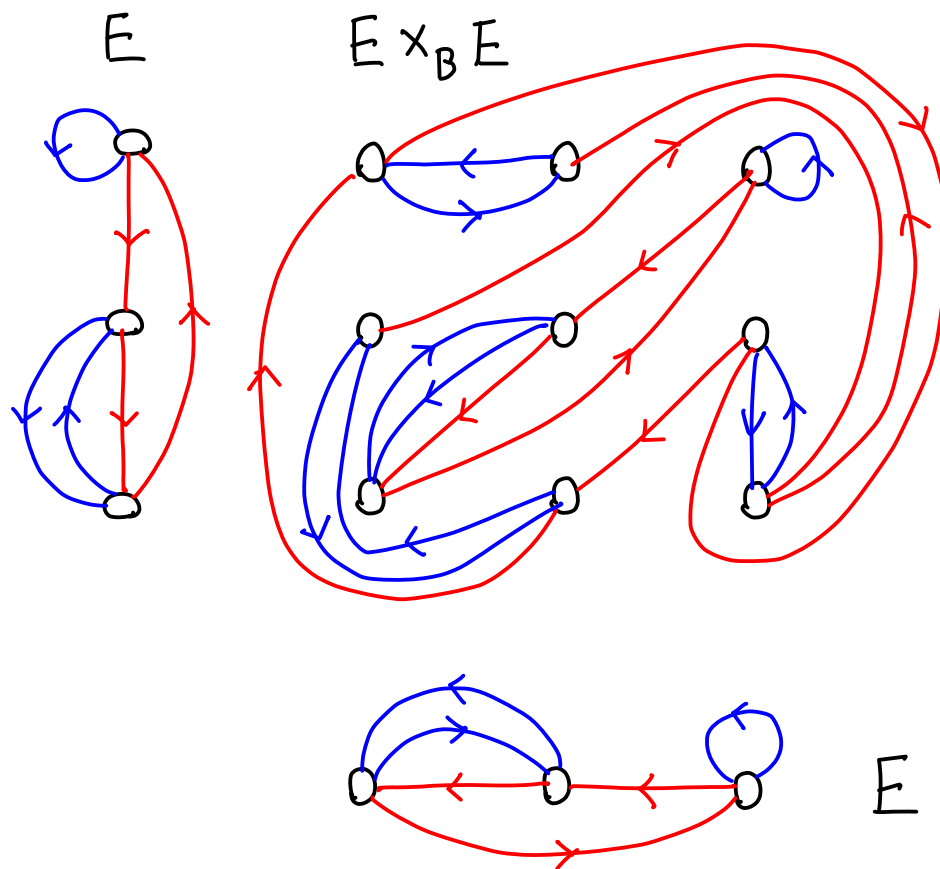
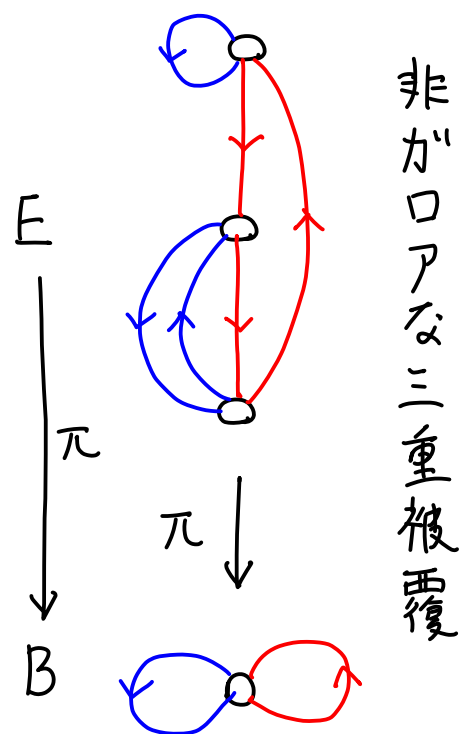
non-Galois

=



Galois 2で割る





$E \times_B E$ は次の2つの連結成分に分かれている:

